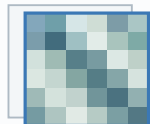


(a) Multimodal segmentation model

IMAGE BRANCH

ResNet-18 U-Net · ImageNet init.



Sentinel-2 RGB · $128 \times 128 \times 3$

PHOTON BRANCH

10 ICESat-2 features



MLP · ELU + dropout



broadcast
 $128 \times 128 \times 16$

image map
 $128 \times 128 \times 16$

concatenate
32 channels

squeeze-
excitation

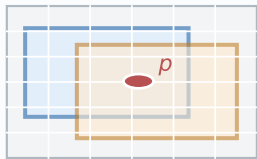
3×3 conv
BN + ReLU
dropout
 1×1 conv



3-class mask

(b) Crop-alignment audit

shared weights · no retraining



overlapping crops c, c'

same trained
model f

prediction at p
 $f(p | c)$

own label $L_c(p)$

other labels $L_{c'}(p)$

$\kappa = \mathbb{E}[\text{own} - \text{other}]$

$\kappa = 0$ if predictions are
crop-invariant